Prevalence of Pathogenic Yersinia enterocolitica Strains in Pigs in the United States

Yersinia enterocolitica is considered an important food-borne pathogen impacting the pork production and processing industry in the United States. Since this bacterium is a commensal of swine, the primary goal of this study was to determine the prevalence of pathogenic Y. enterocolitica in pigs in the United Sates using feces as the sample source. A total of 2,793 fecal samples were tested for its presence in swine. Fecal samples were collected from late finisher pigs from 77 production sites in the 15 eastern and midwestern pork-producing states over a period of 27 weeks (6 September 2000 to 20 March 2001). The prevalence of ail-positive Y. enterocolitica was determined in samples using both a fluorogenic 5′ nuclease PCR assay and a culture method. The mean prevalence was 13.10% (366 of 2,793 fecal samples tested) when both PCR- and culture-positive results were combined. Forty-one of 77 premises (53.25%) contained at least one fecal sample positive for the ail sequence. The PCR assay indicated a contamination rate of 12.35% (345/2,793) compared to 4.08% (114/2,793) by the culture method. Of the 345 PCR-positive samples, 252 were culture negative, while of the 114 culture-positive samples, 21 were PCR negative. Among 77 premises, the PCR assay revealed a significantly ($P < 0.05$) higher percentage (46.75%, $n = 36$ sites) of samples positive for the pathogen (ail sequence) than the culture method (22.08%, $n = 17$ sites). Thus, higher sensitivity, with respect to number of samples and sites identified as positive for the PCR method compared with the culture method for detecting pathogenic Y. enterocolitica, was demonstrated in this study. The results support the hypothesis that swine are a reservoir for Y. enterocolitica strains potentially pathogenic for humans.